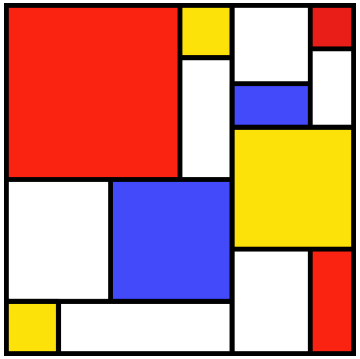
SLiM

Workshop Series



Chromosomes, Haploidy, and Sex

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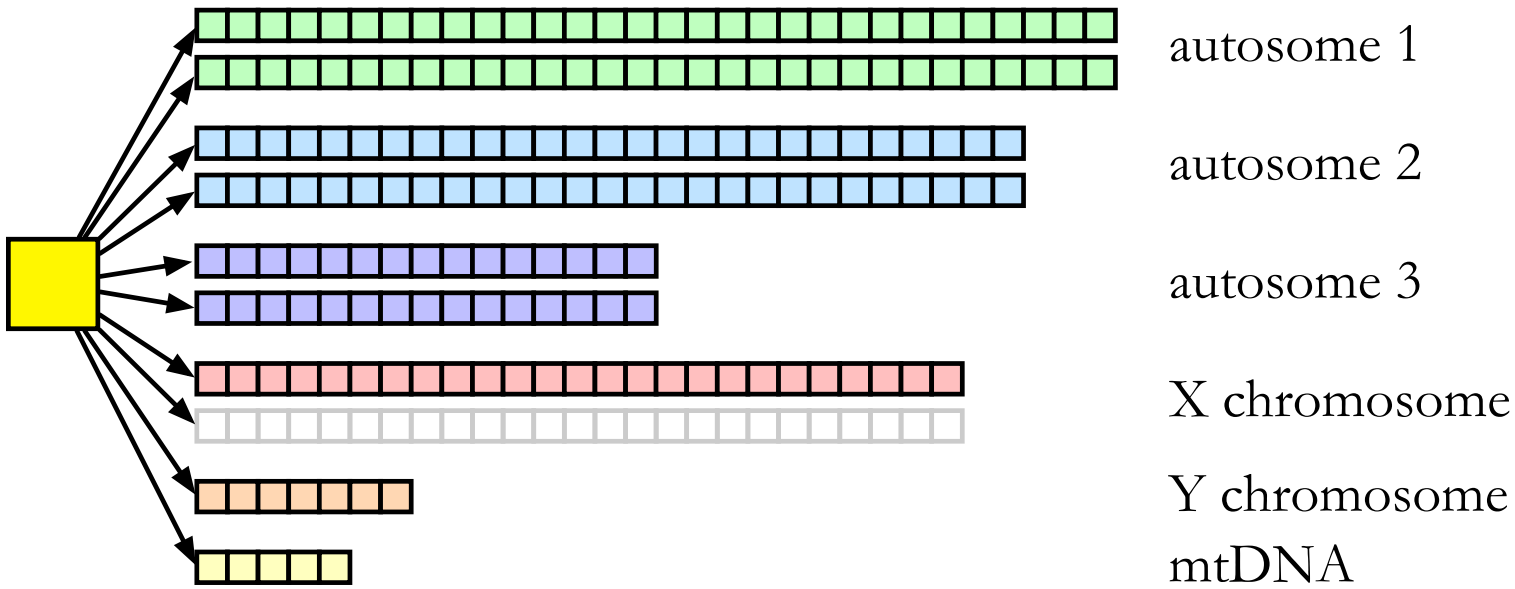
Chromosomes Revisited

- Remember this picture from before:



- How can we set up a model like this?

Chromosomes Revisited

- So far, the chromosome has been *implicit*
- Chromosomes can be declared *explicitly*:
  - `initializeChromosome(...)`
- This has several advantages:
  - You can declare **more than one** chromosome
  - You can specify the **length** up front
  - You can configure the chromosome **type** and other properties (`id`, `symbol`, `name`)

initializeChromosome()

- Explicit chromosome declaration is with:
  - `(object<Chromosome>$)initializeChromosome(integer$ id, [Ni$ length = NULL], [string$ type = "A"], [Ns$ symbol = NULL], ...)`
- We will focus on these options:
  - `id` is an `integer` identifier for the chromosome
  - `length` is its length, in base positions
  - `type` governs its ploidy and inheritance
  - `symbol` is a `string` symbol for the chromosome

A Simple Example

implicit declaration:

```
initialize() {
    initializeMutationType("m1", 0.5, "f", 0.0);
    initializeGenomicElementType("g1", m1, 1.0);

    initializeGenomicElement(g1, 0, 99999);
    initializeMutationRate(1e-7);
    initializeRecombinationRate(1e-8);
}
```

explicit declaration:

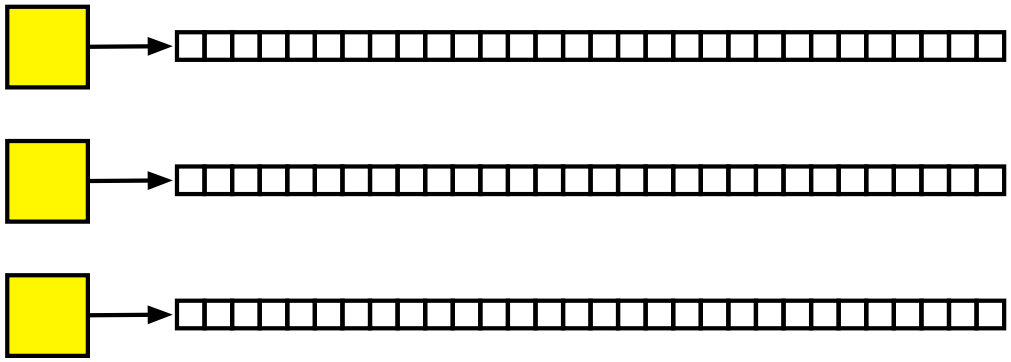
```
initialize() {
    initializeMutationType("m1", 0.5, "f", 0.0);
    initializeGenomicElementType("g1", m1, 1.0);

    initializeChromosome(1, 1e5);
    initializeGenomicElement(g1);
    initializeMutationRate(1e-7);
    initializeRecombinationRate(1e-8);
}
```

Chromosome Types

- Twelve chromosome types are supported
- We will focus on these four types:
  - `"A"` : diploid autosome (the default)
  - `"H"` : haploid autosome
  - `"X"`: an X sex chromosome
  - `"Y"`: a Y sex chromosome
- There are also:
  - `"Z"` and `"W"` for ZW sex chromosomes
  - Several other specialized types

Modeling Haploids

- Modeling haploids is much easier in SLiM 5
- Just use chromosome type `"H"`:
  - `initializeChromosome(1, type="H")`
- This provides an haploid chromosome: one haplosome per individual



Modeling Haploids

- Type `"H"` works in WF and nonWF models
- Usually clonal reproduction (e.g., bacteria)
- Sexual reproduction, with recombination, is also supported (e.g., bryophytes)
- `addRecombinant()` provides full control over reproduction dynamics if needed
- Fitness of mutations is always $1+s$; no dominance coefficient is used

Separate Sexes

- **Hermaphroditic** models are the default
- **Separate sexes** can be modeled easily:
  - Enable with `initializeSex()`
  - Test sex with `individual.sex` (`"M"` or `"F"`)
  - Sex-specific recombination/mutation rates
  - Other support for sex (sampling, output)
  - Sex chromosomes can then be modeled

Separate Sexes in WF

- Sex ratio is a top-down parameter
- The sex ratio is expressed as (M:M+F):
  - `sim.addSubpop("p1", 500, 0.6)`
  - `p1.setSexRatio(0.6)`
- The first parent is always female, the second parent is always male
- Selfing is not allowed

Separate Sexes in nonWF

- Sex-specific `reproduction()` callbacks:
  - `reproduction(<subpop>, "F")`
- Offspring sex can be controlled:
  - `p1.addCrossed(f, m, sex="M")`
  - `p1.addCrossed(f, m, sex=0.6)`
- The sex ratio is emergent from births and deaths, not a top-down parameter

Modeling Sex Chromosomes

- Use chromosome types "X" and "Y"

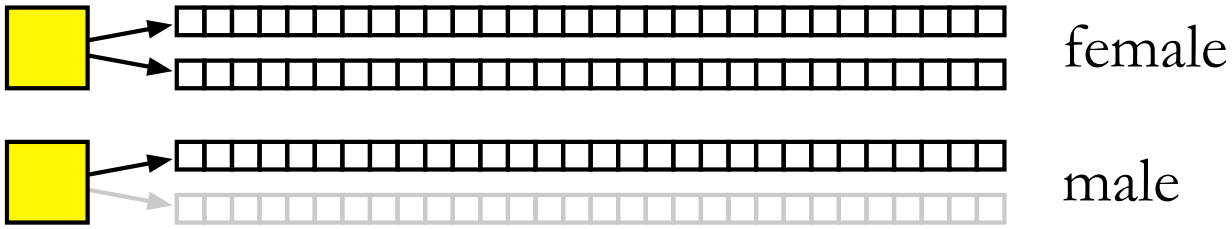
```
initialize() {
    defineConstant("X_LEN", 156040895);
    defineConstant("Y_LEN", 57227415);

    initializeSex();
    initializeMutationType("m1", 0.5, "f", 0.0);
    initializeGenomicElementType("g1", m1, 1.0);

    initializeChromosome(1, X_LEN, type="X", symbol="X");
    initializeGenomicElement(g1);
    initializeMutationRate(1e-7);
    initializeRecombinationRate(1e-8);

    initializeChromosome(2, Y_LEN, type="Y", symbol="Y");
    initializeGenomicElement(g1);
    initializeMutationRate(1e-7);
}
```

Null Haplosomes

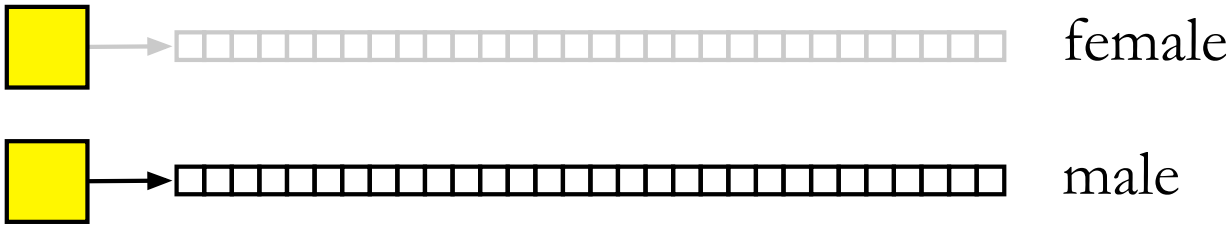
- In SLiM, a chromosome is either:
  - *intrinsically diploid* (two haplosomes)
  - *intrinsically haploid* (one haplosome)
- But sometimes you don't want that!
  - a male has only one X chromosome
  - a female has no Y chromosome at all
  - also cases like haplodiploidy, etc.
- A **null haplosome** is a placeholder

Modeling Sex Chromosomes

- The "X" type is intrinsically diploid; in males, the second haplosome is null:



- The "Y" type is intrinsically haploid; in females, the haplosome is null:

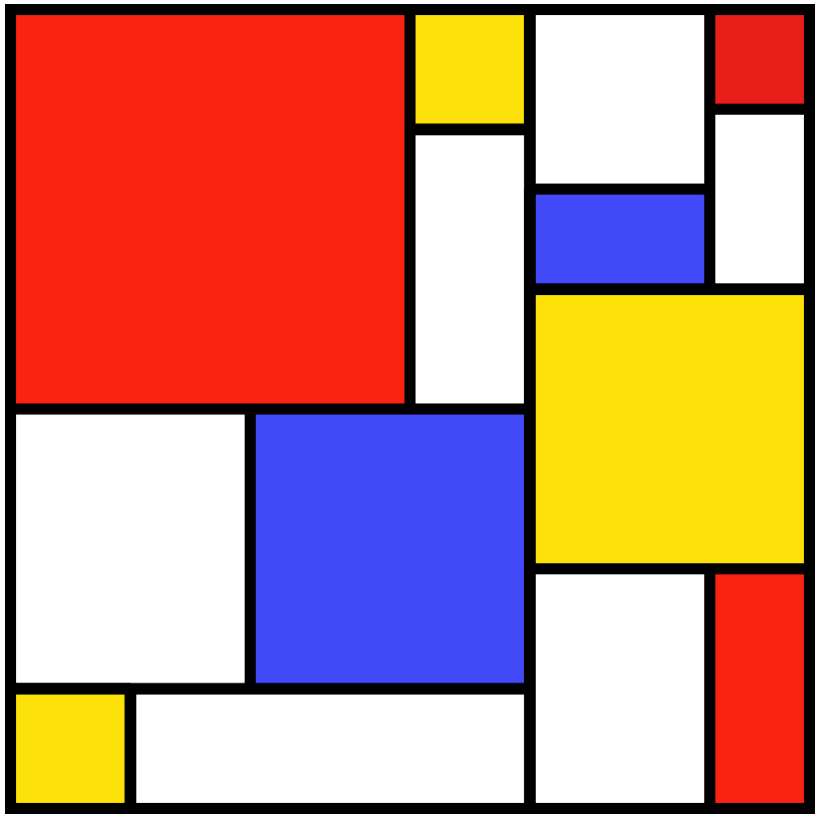


Null Haplosomes

- A given chromosome always uses the same number of haplosomes per individual (its *intrinsic ploidy*):
  - two for `"A"`, one for `"H"`
  - two for `"X"`, one for `"Y"`
- Null haplosomes are placeholders for haplosomes that are not used
  - test with `haplosome.isNullHaplosome`
  - do not attempt to manipulate these!

Modeling Sex Chromosomes

- Many models can be built with this:
  - models with both an X and Y
  - sex chromosomes + autosomes
  - mitochondrial + nuclear DNA
- Some models require further scripting (haplodiploidy, alternation of generations):
  - often, do not use `initializeSex()`
  - control mating, etc., yourself (nonWF)
  - use `addRecombinant()` to generate offspring



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